Linear Algebra and Its Applications

Section 4.7: Change of Basis

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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4.7 Change of basis

A vector stays the same when a basis changes; its list of coordinates changes. We use $P_{C\leftarrow B}$ to mean *from B-coordinates to C-coordinates*:

$$[x]_C = P_{C\leftarrow B}[x]_B, \qquad P_{C\leftarrow B} = [\,[b_1]_C \ \cdots \ [b_n]_C\,].$$

In $\mathbb{R}^n$, if B_0, C_0 are the matrices of the basis vectors in standard coordinates, then $P_{C\leftarrow B} = C_0^{-1}B_0$. In practice reduce $[\,C_0 \ B_0\,]$ to $[\,I \ P_{C\leftarrow B}\,]$. Reversing the change uses the inverse matrix; swapping the labels without inverting is incorrect.

Practice problem 4.7.P1: Read the coordinate labels on the columns

The columns of P are $[f_1]_G, [f_2]_G$. If $v = af_1+bf_2$, linearity of coordinates gives $[v]_G = a[f_1]_G+b[f_2]_G = P(a,b)^T = P[v]_F$. Thus the correct choice is **(ii)**.

Practice problem 4.7.P2: Reverse the change from the example

Example 1 gives $P_{C\leftarrow B} = \left(\begin{smallmatrix} 4 & -6 \\ 1 & 1 \end{smallmatrix}\right)$. Its determinant is $4 + 6 = 10$, so the inverse is

$$P_{B\leftarrow C} = \frac{1}{10}\begin{pmatrix} 1 & 6 \\ -1 & 4 \end{pmatrix}.$$

Multiplying it by the original matrix gives I_2, confirming that changing there and back restores every coordinate vector.

Exercise 4.7.1: Construct columns directly from the given equations

(a) Record the C-coordinates of each b-vector.

From $b_1 = 6c_1 - 2c_2$ and $b_2 = 9c_1 - 4c_2$,

$$P_{C\leftarrow B} = \begin{pmatrix} 6 & 9 \\ -2 & -4 \end{pmatrix}.$$

(b) Apply that change to the specified vector.

For $x = -3b_1 + 2b_2$, its B-coordinate vector is $(-3, 2)$, so

$$[x]_C = \begin{pmatrix} 6 & 9 \\ -2 & -4 \end{pmatrix}\begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} -18+18 \\ 6-8 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}.$$

Exercise 4.7.2: A singular matrix reveals a printed inconsistency

The supplied relations are $b_1 = -2c_1 + 4c_2$ and $b_2 = 3c_1 - 6c_2$. But then $b_2 = -\frac{3}{2}b_1$, so $\{b_1, b_2\}$ is dependent and cannot be a basis. These relations are incompatible with the printed assumption that B and C are both bases. This is visible in the formal column matrix

$$Q = \begin{pmatrix} -2 & 3 \\ 4 & -6 \end{pmatrix}, \qquad \det Q = 12 - 12 = 0.$$

(a) State what the formal calculation does and does not mean.

The columns are the given C-coordinates, but Q is not a change-of-basis matrix between two bases and has no inverse. There is no valid instance satisfying all the printed assumptions.

(b) Evaluate the specified linear combination nevertheless.

The vector $x = 2b_1 + 3b_2$ is meaningful even for dependent generators:

$$x = (-4 + 9)c_1 + (8 - 18)c_2 = 5c_1 - 10c_2.$$

Thus $[x]_C = (5, -10)$. The coefficient list $(2, 3)$ is one representation using b_1, b_2, but cannot be called a unique B-coordinate vector.

Exercise 4.7.3: The output coordinate system matches the columns

The columns are $[u_1]_W, [u_2]_W$, so multiplying by the coefficients of x in U forms the W-coordinates of x. Hence $[x]_W = P[x]_U$, choice **(ii)**. The reverse equation would need P^{-1}.

Exercise 4.7.4: A three-vector version of the same rule

Here P has columns $[d_1]_A, [d_2]_A, [d_3]_A$. Therefore the input coefficients are D-coordinates, and the output is an A-coordinate vector: $[x]_A = P[x]_D$. The correct choice is **(i)**.

Exercise 4.7.5: Translate three linear-combination equations

(a) Put the B-coordinates of a_1, a_2, a_3 into columns.

The equations give

$$P_{B \leftarrow A} = \begin{pmatrix} 4 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

(b) Multiply by the A-coordinate vector.

For $x = 3a_1 + 4a_2 + a_3$,

$$[x]_B = \begin{pmatrix} 4 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & -2 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 2 \\ 2 \end{pmatrix}.$$

For example, the first output entry is $4(3) - 1(4) + 0(1) = 8$.

Exercise 4.7.6: Change from F to D

The columns must be $[f_1]_D, [f_2]_D, [f_3]_D$, giving

$$P_{D \leftarrow F} = \begin{pmatrix} 2 & 0 & -3 \\ -1 & 3 & 0 \\ 1 & 1 & 2 \end{pmatrix}.$$

For $x = f_1 - 2f_2 + 2f_3$, multiply by $(1, -2, 2)$ to obtain

$$[x]_D = (2 - 6,\ -1 - 6,\ 1 - 2 + 4) = (-4, -7, 3).$$

This answers (a) and (b), keeping the direction of the coordinate change explicit.

Exercise 4.7.7: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} 7 & -3 \\ 5 & -1 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 1 & -2 \\ -5 & 2 \end{pmatrix}$. Solve $C_0 P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C \leftarrow B} = \begin{pmatrix} -3 & 1 \\ -5 & 2 \end{pmatrix}, \qquad P_{B \leftarrow C} = \begin{pmatrix} -2 & 1 \\ -5 & 3 \end{pmatrix}.$$

The first identity is checked by $C_0 P_{C \leftarrow B} = B_0$; the second is the inverse, checked by $P_{B \leftarrow C} P_{C \leftarrow B} = I_2$. For example, column one of $C_0 P$ reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.8: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} -1 & 1 \\ 8 & -7 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$. Solve $C_0 P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C \leftarrow B} = \begin{pmatrix} 9 & -8 \\ -10 & 9 \end{pmatrix}, \qquad P_{B \leftarrow C} = \begin{pmatrix} 9 & 8 \\ 10 & 9 \end{pmatrix}.$$

The first identity is checked by $C_0 P_{C \leftarrow B} = B_0$; the second is the inverse, checked by $P_{B \leftarrow C} P_{C \leftarrow B} = I_2$. For example, column one of $C_0 P$ reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.9: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} 4 & 8 \\ 4 & 4 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix}$. Solve $C_0 P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C \leftarrow B} = \begin{pmatrix} 2 & 3 \\ 0 & -1 \end{pmatrix}, \qquad P_{B \leftarrow C} = \begin{pmatrix} \frac{1}{2} & \frac{3}{2} \\ 0 & -1 \end{pmatrix}.$$

The first identity is checked by $C_0 P_{C \leftarrow B} = B_0$; the second is the inverse, checked by $P_{B \leftarrow C} P_{C \leftarrow B} = I_2$. For example, column one of $C_0 P$ reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.10: Compute both directions in standard coordinates

The standard-coordinate basis matrices are $B_0 = \begin{pmatrix} 6 & 4 \\ -12 & 2 \end{pmatrix}$ and $C_0 = \begin{pmatrix} 4 & 3 \\ 2 & 9 \end{pmatrix}$. Solve $C_0P = B_0$ column by column, or reduce $[\,C_0\ B_0\,]$, to obtain

$$P_{C\leftarrow B} = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}, \qquad P_{B\leftarrow C} = \begin{pmatrix} 0 & -\frac{1}{2} \\ 1 & \frac{3}{2} \end{pmatrix}.$$

The first identity is checked by $C_0P_{C\leftarrow B} = B_0$; the second is the inverse, checked by $P_{B\leftarrow C}P_{C\leftarrow B} = I_2$. For example, column one of C_0P reconstructs b_1 as a combination of the c-vectors. These checks ensure that the two directions have not been interchanged.

Exercise 4.7.11: Two statements about the direction of a change

(a) False.

The columns of $P_{C\leftarrow B}$ are the C-coordinates of the B vectors, $[b_j]_C$, not the B-coordinates of the C vectors. The latter columns belong to its inverse $P_{B\leftarrow C}$.

(b) True.

If C is the standard basis of $\mathbb{R}^n$, $[x]_C = x$ and $[b_j]_C = b_j$. Thus $P_{C\leftarrow B} = [\,b_1\ \cdots\ b_n\,] = P_B$.

Exercise 4.7.12: Independence and an algorithm's direction

(a) True.

A change-of-coordinates matrix between bases is invertible: composing it with the reverse change gives identity. By the Invertible Matrix Theorem, its columns are therefore independent.

(b) False.

Reducing $[\,c_1\ c_2\ b_1\ b_2\,]$ to $[\,I\ P\,]$ computes $P = C_0^{-1}B_0$. It therefore satisfies $[x]_C = P[x]_B$, the reverse of the printed assertion. The printed equation would use P^{-1} instead.

Exercise 4.7.13: Change from a polynomial basis to coefficients

The standard coordinate vector of $a + bt + ct^2$ is (a, b, c). Consequently, reading coefficients of the three given polynomials gives

$$P_{C\leftarrow B} = \begin{pmatrix} 1 & 3 & 0 \\ -2 & -5 & 2 \\ 1 & 4 & 3 \end{pmatrix}.$$

To find $[-1 + 2t]_B$, solve $Pz = (-1, 2, 0)$. Exact elimination gives $z = (5, -2, 1)$. Check directly:

$$5(1 - 2t + t^2) - 2(3 - 5t + 4t^2) + (2t + 3t^2) = -1 + 2t.$$

Thus the coordinate vector is $(5, -2, 1)$, and the displayed matrix is the requested change to the standard basis.

Exercise 4.7.14: Represent t^2 using three different polynomials

Reading constant, linear, and quadratic coefficients gives

$$P = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ -3 & -5 & 0 \end{pmatrix}.$$

Solve $Pz = (0, 0, 1)$. Its first two equations are $z_1 + 2z_2 + z_3 = 0$ and $z_2 + 2z_3 = 0$, so $z_2 = -2z_3$ and $z_1 = 3z_3$. The last equation gives $-9z_3 + 10z_3 = 1$, hence $z_3 = 1$. Therefore

$$t^2 = 3(1 - 3t^2) - 2(2 + t - 5t^2) + (1 + 2t).$$

The coefficients $(3, -2, 1)$ can be checked by canceling constant and linear terms.

Exercise 4.7.15: Justify the construction of the change matrix

(a) B spans V, because B is a basis. Therefore v has an expansion $v = \sum_j x_j b_j$.

(b) The coordinate mapping determined by a basis is linear. It takes that expansion to $[v]_C = \sum_j x_j [b_j]_C$.

(c) This is the definition of a matrix-vector product as a linear combination of the matrix's columns.

(d) The column $(x_1, \ldots, x_n)$ is $[v]_B$, by the definition and uniqueness of coordinates relative to B.

These observations prove $[v]_C = P_{C \leftarrow B}[v]_B$ for every vector v.

Exercise 4.7.16: Justify uniqueness of the change matrix

(a) $[b_1]_B = e_1$, so $Q[b_1]_B = Qe_1$ is the first column of Q.

(b) For a general k, column k must be $[b_k]_C$.

(c) Indeed $[b_k]_B = e_k$, and multiplying Q by e_k selects its kth column.

Thus every column of Q is forced. It must equal $[\,[b_1]_C \;\cdots\; [b_n]_C\,] = P_{C \leftarrow B}$.

Exercise 4.7.17: Change between powers of cosine and multiple angles

Write $x_j(t) = \cos^j t$, $y_j(t) = \cos(jt)$ for $j = 0, \ldots, 6$.

(a) Form P from the multiple-angle identities, then invert it.

The angle-addition formulas recursively give the polynomials $T_0(u) = 1$, $T_1(u) = u$, $T_{j+1}(u) =$

$2uT_j(u) - T_{j-1}(u)$ with $y_j(t) = T_j(\cos t)$. Taking their coefficients as columns gives

$$P = \begin{pmatrix} 1 & 0 & -1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & -3 & 0 & 5 & 0 \\ 0 & 0 & 2 & 0 & -8 & 0 & 18 \\ 0 & 0 & 0 & 4 & 0 & -20 & 0 \\ 0 & 0 & 0 & 0 & 8 & 0 & -48 \\ 0 & 0 & 0 & 0 & 0 & 16 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 32 \end{pmatrix}.$$

$$P^{-1} = \begin{pmatrix} 1 & 0 & \frac{1}{2} & 0 & \frac{3}{8} & 0 & \frac{5}{16} \\ 0 & 1 & 0 & \frac{3}{4} & 0 & \frac{5}{8} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & \frac{15}{32} \\ 0 & 0 & 0 & \frac{1}{4} & 0 & \frac{5}{16} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & \frac{3}{16} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{32} \end{pmatrix}.$$

Exact multiplication gives $PP^{-1} = P^{-1}P = I_7$.

(b) Interpret every column, rather than just listing numbers.

Since $P = P_{B \leftarrow C}$, its inverse converts B-coordinates to C-coordinates. The jth x-vector has B-coordinates equal to the corresponding standard coordinate vector, so its C-coordinates are a column of P^{-1}. Reading those seven columns gives

$$\begin{aligned} 1 &= \cos 0t, \\ \cos t &= \cos t, \\ \cos^2 t &= \tfrac{1}{2} + \tfrac{1}{2}\cos 2t, \\ \cos^3 t &= \tfrac{3}{4}\cos t + \tfrac{1}{4}\cos 3t, \\ \cos^4 t &= \tfrac{3}{8} + \tfrac{1}{2}\cos 2t + \tfrac{1}{8}\cos 4t, \\ \cos^5 t &= \tfrac{5}{8}\cos t + \tfrac{5}{16}\cos 3t + \tfrac{1}{16}\cos 5t, \\ \cos^6 t &= \tfrac{5}{16} + \tfrac{15}{32}\cos 2t + \tfrac{3}{16}\cos 4t + \tfrac{1}{32}\cos 6t. \end{aligned}$$

Exercise 4.7.18: Integrate after changing the function basis

The integrand has B-coordinates $(0, 0, 0, 5, -6, 5, -12)$ in the notation of Exercise 4.7.17. Multiplication by P^{-1} gives its C-coordinates

$$(-6,\ 55/8,\ -69/8,\ 45/16,\ -3,\ 5/16,\ -3/8).$$

Thus it equals

$$-6 + \tfrac{55}{8}\cos t - \tfrac{69}{8}\cos 2t + \tfrac{45}{16}\cos 3t - 3\cos 4t + \tfrac{5}{16}\cos 5t - \tfrac{3}{8}\cos 6t.$$

Integrate these simple cosine terms using $\int \cos(jt)\,dt = \sin(jt)/j$ for $j > 0$:

$$\int (5\cos^3 t - 6\cos^4 t + 5\cos^5 t - 12\cos^6 t)\,dt$$

$$= -6t + \tfrac{55}{8}\sin t - \tfrac{69}{16}\sin 2t + \tfrac{15}{16}\sin 3t - \tfrac{3}{4}\sin 4t + \tfrac{1}{16}\sin 5t - \tfrac{1}{16}\sin 6t + C.$$

Differentiating the result restores the multiple-angle expression, which Exercise 4.7.17 identifies with the original polynomial in cosine.

Exercise 4.7.19: Find bases when a change matrix is prescribed

Let $V_0 = [\, v_1 \; v_2 \; v_3 \,]$ and let P be the supplied matrix.

(a) Its columns give the new u-vectors in v-coordinates.

For $P = P_{V \leftarrow U}$ we need $U_0 = V_0 P$. Multiplication gives

$$U_0 = \begin{pmatrix} -6 & -6 & -5 \\ -5 & -9 & 0 \\ 21 & 32 & 3 \end{pmatrix}.$$

Take its columns as u_1, u_2, u_3. Since V_0 and P are invertible (the latter has determinant -1), their product is invertible, so these columns are a basis.

(b) Solve the reversed requirement.

Now $P = P_{W \leftarrow V}$ means $V_0 = W_0 P$, so $W_0 = V_0 P^{-1}$. This gives

$$W_0 = \begin{pmatrix} 28 & 38 & 21 \\ -9 & -13 & -7 \\ -3 & 2 & 3 \end{pmatrix}.$$

Take these three columns as w_1, w_2, w_3. Their matrix is invertible for the same reason, and $W_0 P = V_0$ checks the required direction.

Exercise 4.7.20: Compose changes in the same order as their actions

(a) Apply the first change, then the second.

For every $x \in V$,

$$[x]_D = P_{D \leftarrow C}[x]_C = P_{D \leftarrow C} P_{C \leftarrow B}[x]_B.$$

By uniqueness of the change matrix, $P_{D \leftarrow B} = P_{D \leftarrow C} P_{C \leftarrow B}$. The rightmost factor acts first, exactly as in composition of functions.

(b) A concrete exact check.

Use the bases with standard-coordinate matrices

$$B_0 = \begin{pmatrix} 7 & -3 \\ 5 & -1 \end{pmatrix}, \quad C_0 = \begin{pmatrix} 1 & -2 \\ -5 & 2 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -1 & 1 \\ 8 & -7 \end{pmatrix}.$$

The program calculation $C_0^{-1}B_0$, $D_0^{-1}C_0$, $D_0^{-1}B_0$ (using exact matrix inversion) gives

$$\begin{pmatrix} 2 & -12 \\ 3 & -14 \end{pmatrix} \begin{pmatrix} -3 & 1 \\ -5 & 2 \end{pmatrix} = \begin{pmatrix} 54 & -22 \\ 61 & -25 \end{pmatrix}.$$

The right side is $D_0^{-1}B_0$, independently verifying the composition rule.